

Unusual Does Not Mean Impossible

AP Statistics · Unit 2 · Topic 2.10 · B: 2026-11-06 · E: 2026-11-30 · TEACHER KEY

CED TETHER

- **Skill 3.B** — Using Probability and Simulation: Determine parameters for probability distributions.
- **Skill 4.B** — Statistical Argumentation: Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.C** — Calculate parameters for a binomial distribution. [Skill 3.B]
- **EK UNC-3.C.1** — If a random variable is binomial, its mean, μ_X , is np and its standard deviation, σ_X , is $\sqrt{np(1-p)}$.
- **LO UNC-3.D** — Interpret probabilities and parameters for a binomial distribution. [Skill 4.B]
- **EK UNC-3.D.1** — Probabilities and parameters for a binomial distribution should be interpreted using appropriate units and within the context of a specific population or situation.

Essential question: How do binomial parameters help distinguish an unusual count from an impossible one?

DOK-3 skill: 4.B · **FRQ pattern:** binomial-mean-sd-unusual-vs-impossible

DOK rationale: Part (c) coordinates binomial parameters, a standardized distance, approximation conditions, and competing meanings of rare and impossible to assess a coach's claim and its consequence.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask what numerical boundary, if any, the word impossible should imply.
- Begin the combined 4.10–4.12 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and connect parameter calculations to claim language.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose the coach or analyst and name one quantity to check before the lesson.
- Complete the follow-along about binomial mean, standard deviation, and interpretation.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What does 70 describe across many 100-shot sets, and what does it not guarantee?
- How many standard deviations below 70 is 58?
- Do both expected-count checks support a normal approximation?
- What probability would the word impossible require?

Adult role

Solo teacher. Circulate 5 pairs. Ask students to attach makes to every parameter and to distinguish distance from the mean from probability zero.

[WARN] WATCH FOR

- Reporting 0.70 rather than 70 as the mean of the count.
- Omitting the square root or the failure probability from the standard deviation.
- Treating a two-standard-deviation guideline as a boundary beyond which outcomes cannot occur.

SCORING PART (c) — E / P / I

Expected elements

- **computes-parameters** (required) — Computes $\mu_X = 70$ and $\sigma_X = \sqrt{21} \approx 4.6$ makes
- **standardizes-and-checks-normality** (required) — Computes $z \approx -2.6$ and verifies $np = 70$ and $n(1 - p) = 30$ are both at least 10
- **distinguishes-unusual-from-impossible** (required) — Calls 58 unusual while explaining that a low-probability binomial outcome still has positive probability
- **states-misleading-consequence** (required) — Explains that the coach treats the mean or typical range as a guaranteed cutoff and overclaims from one set

E	Correctly computes and contextualizes the parameters and z-score, verifies both approximation counts, supports the analyst, and clearly explains both the rare-versus-zero distinction and the coach's misleading cutoff.
P	Correctly identifies 58 as unusual with most calculations but omits an approximation count, the rare-versus-zero mechanism, or the specific consequence of the coach's statement.
I	Uses 0.70 as the mean count, omits the square root in the standard deviation, calls 58 impossible, or treats the mean as a guaranteed result.

Common mistakes

- Using $\sqrt{np}$ instead of $\sqrt{np(1 - p)}$
- Dividing 58 by the standard deviation instead of subtracting the mean first
- Checking only np and not $n(1 - p)$
- Equating more than two standard deviations from the mean with zero probability

[STAR] TODAY'S PROBLEM — annotated

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: “Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies.”

Analyst: “A result can be unusual without being impossible; compare 58 with the model’s mean and standard deviation.”

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE (5 min)

Whose statement is more defensible, the coach’s or the analyst’s? Name one quantity you would check.

Key: Not graded. Listen for whether students use impossible to mean unlikely; the calculation should sharpen that language.

Rules callout “CENTER, SPREAD, AND UNUSUAL COUNTS (keep on the page)” is printed on the student sheet.

DOK 1 (a) State n and p for X and write the formulas for the binomial mean and standard deviation.

Key: Here $n = 100$ and $p = 0.70$. For a binomial count, $\mu_X = np$ and $\sigma_X = \sqrt{np(1-p)}$.

DOK 2 (b) Without calculating yet, interpret the model’s mean and standard deviation in the context of many sets of 100 free throws.

Key: Across many comparable sets of 100 shots under the 0.70 model, the number made would average μ_X makes. A set’s number made would typically differ from that mean by about σ_X makes; neither parameter guarantees the result in one set.

DOK 3 (c) Compute μ_X , σ_X , and the z -score of 58. Check np and $n(1-p)$ to decide whether a normal approximation is reasonable. Is 58 unusual? Adjudicate the two statements, distinguish “unusual” from “impossible,” and explain what the coach’s claim could mislead a reader into believing.

Key: $\mu_X = 100(0.70) = 70$ makes and $\sigma_X = \sqrt{100(0.70)(0.30)} = \sqrt{21} = 4.5826 \approx 4.6$ makes. Thus $z = (58 - 70)/4.5826 = -2.6186 \approx -2.6$. Also, $np = 70$ and $n(1-p) = 30$, both at least 10, so a normal approximation is reasonable. Because 58 is about 2.6 standard deviations below the mean, it is unusual. The analyst is warranted: unusual means low probability under the model, not zero probability. In fact, the exact binomial probability $P(X \leq 58) \approx 0.00717$, and $P(X < 60) \approx 0.01250$, so fewer than 60 remains possible. The coach turns an expected center into a hard lower bound and could mislead a reader into thinking every 70 percent shooter must make at least 60 of every 100 attempts or that one unusual set proves the parameter changed.

EXIT

One thing the video changed about my first take: _____